

(3)

Grouped data is data given in intervals

Ex	Excess Score	Frequency
90-99		7
80-89		5
70-79		15
60-69		4
50-59		5
40-49		0
30-39		1
Total		37

Ex: Calculate the arithmetic mean of the marks from the following table

Marks:	0-10	10-20	20-30	30-40	40-50	50-60
No. of Student:	12	18	27	20	17	6

Marks	No. of Student (f)	Mid pt (x)	fx
0-10	12	5	60
10-20	18	15	270
20-30	27	25	675
30-40	20	35	700
40-50	17	45	765
50-60	6	55	330
Total	N = 100		2800

$$\text{Arithmetic mean } (\bar{x}) = \frac{1}{N} \sum fx = \frac{2800}{100} = 28$$

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(4)

Mean of the composite series

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + \dots + n_k \bar{x}_k}{n_1 + n_2 + \dots + n_k} = \frac{\sum h_i \bar{x}_i}{\sum h_i}$$

Ques The average salary of male employees in a firm was Rs. 5200 and that of female was Rs. 4200. The mean salary of all the employees was Rs. 5000. Find the percentage of male and female employees

Soln

$n_1 \rightarrow$ No. of male emp

$n_2 \rightarrow$ No. of female emp

$\bar{x}_1$ & $\bar{x}_2 \rightarrow$ Average salary in rupees

$\bar{x} \rightarrow$ Average salary of all workers in the firm

$$\bar{x}_1 = 5200, \bar{x}_2 = 4200 \text{ \& } \bar{x} = 5000$$

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$5000(n_1 + n_2) = 5200n_1 + 4200n_2$$

$$200n_1 = 800n_2$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{4}{1}$$

$\therefore$ The percentage of male employees in the firm

$$= \frac{4}{4+1} \times 100 = 80\%$$

$\&$ The percentage of female employees in the firm

$$= \frac{1}{4+1} \times 100 = 20\%$$

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Arithmetic mean - Short-cut method

$$\bar{x} = A + \frac{\sum fd}{N}$$

For discrete series

where $A =$ Assumed mean

$$d = X - A$$

$N =$ Total no. of observations i.e. $\sum f$

Step ① Take an assumed mean

(ii) Take the deviations of variable X from assumed mean ($d = X - A$)

(iii) Multiply deviations with the respective frequency and take the total $\sum fd$

(iv) Divide the total obtained in (iii) by total frequency

Ex

Find the arithmetic mean

X	No. of Student f	(X - A) d	fd
20	8	-20	-160
30	12	-10	-120
40	20	0	0
50	10	10	100
60	6	20	120
70	4	30	120
	N = 60		$\sum fd = 60$

$$\bar{x} = A + \frac{\sum fd}{N} = 40 + \frac{60}{60} = 41$$

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Arithmetic mean - continuous data (Short-cut method)

$$\bar{x} = A + \frac{\sum fd}{N}$$

$A =$ assumed mean, $d =$ deviation of X from assumed mean i.e. $(m - A)$

$N =$ total no. of observations

mid pt.

Ex

Marks	mid-point m	No. of Student f	(m - A) = d	fd
0-10	5	5	-30	-150
10-20	15	10	-20	-200
20-30	25	25	-10	-250
30-40	35 $\rightarrow A$	30	0	0
40-50	45	20	10	200
50-60	55	10	20	200
		$\sum f = N = 100$		$\sum fd = -200$

$$\bar{x} = A + \frac{\sum fd}{N} = 35 + \frac{(-200)}{100} = 33$$

Simplified formula

divide the deviation $(m - A)$ by the class intervals i.e. calculate $(m - A)/h$ and then multiply by f_i

$$\bar{x} = A + \frac{\sum fd}{N} \times h$$

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Ex	Marks	mid point (m)	No. of student (f)	(m-35)	(m-35)/h	fd
	0-10	5	5	-30	-3	-15
	10-20	15	10	-20	-2	-20
	20-30	25	25	-10	-1	-25
	30-40	35 ^{-A}	30	0	0	0
	40-50	45	20	10	1	20
	50-60	55	10	20	2	20
			N=100	Σfd = -20		

$$\bar{x} = A + \frac{\Sigma fd}{N} \times h$$

$$= 35 - \frac{20}{100} \times 10 = 33$$

* If the class of intervals are unequal we can simplify calculations by taking a common factor. In such case we should use $\frac{m-A}{c}$ instead of $\frac{m-A}{h}$ while calculating.

Ex	Marks	Mid-point	f	$\frac{m-45}{c} = d$	fd
	0-10	5	5	-8	-40
	10-30	20	12	-5	-60
	30-60	45 ^{-A}	25	0	0
	60-100	80	8	7	56
			N=50	Σfd = -44	

$$\bar{x} = A + \frac{\Sigma fd}{N} \times c = 45 - \frac{44}{50} \times 5 = 40.6$$

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Merits of Arithmetic mean

- Simplest average to understand and easiest compute.
- It is affected by the value of every item in series.
- It is defined by a rigid formula.
- It is relatively reliable in the sense that it does not vary too much when repeated samples are taken from one and the same population.
- Mean is typical in the sense that it is the centre of gravity, balancing the values on either side of it.
- It is a calculated value, and not based on position in the series.

Limitations

- Since the value of mean depends upon each item of the series, extreme items (a very small and very large items), unduly affect the value of average. The smaller the no. of observations, the greater is likely to be the impact of extreme values.
- In a distribution with open-end classes, mean cannot be computed without making assumptions. If such classes contain a large proportion of the values, then mean may be subject to substantial error.
- Arithmetic mean is not always a good measure of central tendency. The mean provides a characteristic value, in the sense of indicating where most of the values lie, only when the distribution of the variable is reasonably normal.

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Median

Median of a distribution is the value of the variable which divides it into two equal parts.

* Median is a positional average.

Calculation of median - Individual observation

Step - (1) Arrange data in ascending or descending order.

(2) If no. of data is odd then median is the middle value.

If no. of data is even, then arithmetic mean of the middle values is median.

Ex (1) Data of wages of 7 workers (odd)

Wages 14,100, 14,150, 16,080, 17,120, 15,200, 16,160, 17,400

Sol -

Step-1

Sr	Wages (ascending)
1	14,100
2	14,150
3	15,200
4	16,080
5	16,160
6	17,120
7	17,400

Here no. of data is odd

$$\text{Median} = \frac{N+1}{2} = \frac{7+1}{2} = 4^{\text{th}} \text{ item} = 16,080 \text{ Rs}$$

Ex (2) Monthly income of 10 employees (Even)

14,391, 15,384, 15,407, 16,672, 26,522, 16,777, 26,753, 27,850, 37,490

Soln

Sr Income (Ascending order)

1	14,391
2	15,384
3	15,407
4	16,672
5	16,777
6	25,591
7	26,522
8	26,753
9	27,850
10	37,490

$$\text{Median} = \text{Size} + \frac{N+1}{2} \text{th item} = \frac{10+1}{2} = 5.5^{\text{th}} \text{ item}$$

$$\begin{aligned} \text{Size of } 5.5^{\text{th}} \text{ item} &= \frac{5^{\text{th}} \text{ item} + 6^{\text{th}} \text{ item}}{2} \\ &= \frac{16,777 + 25,591}{2} \\ &= \frac{42,368}{2} = 21,184 \end{aligned}$$

Computation of Median - Discrete Series

Step (1) Arrange data in ascending or descending order.

(2) Calculate median = Size of $\frac{N+1}{2}$

Find the cumulative freq.

(4) Look at the cumulative freq. and find that total which either equal to $\frac{N+1}{2}$ or next higher to that and determine the value of the corresponding variable.

Income	15,000	15,500	16,800	18,000	18,500	17,800
No. of persons	24	26	16	20	6	30

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Income ascending order	No. of person (f)	Cumulative freq
15000	24	24
15500	26	50
16800	20	70
17800	30	100
18000	16	116
18500	6	122

$$\text{Median} = \text{Size of } \frac{N+1}{2} \text{th item} = \frac{122+1}{2} = 61.5^{\text{th}} \text{ item}$$

$$\text{Size of } 61.5^{\text{th}} \text{ item} = 16800 \text{ Rs}$$

Calculation of Median - Continuous Series

$$\text{Median} = L + \frac{h}{f} \left(\frac{N}{2} - Cf \right)$$

L = lower limit of the median class i.e. the class in which the middle item of the distribution lies.
 Cf = cumulative freq. of the class preceding the median class or sum of the freq. of all classes lower than the median class.

f = Simple freq. of the median class

h = the class interval of the median class

$$N = \sum f$$

Ex Find the median marks

Marks	45-50	40-45	35-40	30-35	25-30
No. of Student	10	15	26	30	42

Marks	20-25	15-20	10-15	5-10
No. of Student	31	24	15	7

Soln First arrange the data in ascending order

Marks	No. of Student (f)	Cf
5-10	7	7
10-15	15	22
15-20	24	46
20-25	31	77
25-30	42	119
30-35	30	149
35-40	26	175
40-45	15	190
45-50	10	200

$$\text{Median} = \text{Size of } \frac{N}{2} \text{th item} = \frac{200}{2} = 100^{\text{th}} \text{ item}$$

Median lies in the class 25-30

$$\text{Median} = L + \frac{h}{f} \left(\frac{N}{2} - Cf \right)$$

$$L = 25, h = 5, f = 42$$

$$N = 200, Cf = 77$$

$$\text{Median} = 25 + \frac{5}{42} \left(\frac{200}{2} - 77 \right)$$

$$= 25 + \frac{5}{42} (23)$$

$$= 27.74$$

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Q Calculate median for the following data

Weight (gm)	No. of Apples	Weight (gm)	No. of Apples
410-419	14	450-459	45
420-429	20	460-469	18
430-439	42	470-479	7
440-449	54		

Since the data given inclusive class interval, we should first convert it into exclusive by deducting 0.5 from lower limit and adding 0.5 to upper limit

Weight	f	Cf	
409.5 - 419.5	14	14	Med = Size of $\frac{N}{2}$ th item
419.5 - 429.5	20	34	$= \frac{200}{2} = 100^{\text{th}}$
429.5 - 439.5	42	76	$\Rightarrow$ Median lies in the
439.5 - 449.5	54	130	Class 439.5 - 449.5
449.5 - 459.5	45	175	$L = 439.5, h = 10$
459.5 - 469.5	18	193	$Cf = 76, f = 54$
469.5 - 479.5	7	200	

$$N = 200$$

$$\text{Med} = L + \frac{h}{f} \left(\frac{N}{2} - Cf \right)$$

$$= 439.5 + \frac{10}{54} \left(\frac{200}{2} - 76 \right)$$

$$= 439.5 + 4.44$$

$$= 443.94$$

Ex. An incomplete distribution is given as

Variable	0-10	10-20	20-30	30-40	40-50	50-60
Freq	10	20	?	40	?	25

Variable	60-70
Freq	15

Find out missing freq. ?

Total freq. is 170 (Given)
 Median " 35 (Given)

Soln Let missing freq. of class 20-30 is f_1
 & " " " " 40-50 is f_2

$$\Rightarrow 10 + 20 + f_1 + 40 + f_2 + 25 + 15 = 170$$

$$\Rightarrow f_1 + f_2 = 60$$

$$\text{Median} = L + \frac{h}{f} \left(\frac{N}{2} - Cf \right)$$

$$\text{Median} = \text{Size of } \frac{N}{2} \text{th item} = \frac{170}{2} = 85^{\text{th}} \text{ item}$$

Median is 35 which lies in the class 30-40

$$\Rightarrow L = 30, \frac{N}{2} = 85, Cf = 10 + 20 + f_1, h = 10$$

$$f = 40$$

$$35 = 30 + \frac{10}{40} (85 - 10 - 20 - f_1)$$

$$f_1 = 35$$

$$\Rightarrow f_2 = 60 - 35$$

$$= 25$$

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Merits of Median

- ① Useful for the open-end classes also recommended if distribution has unequal classes
- ② Extreme values do not affect the median as strongly they do the mean.
- ③ In markedly skewed distributions where the arithmetic mean would be distorted by extreme values, the median is especially useful.
- ④ most appropriate average in dealing with qualitative data.
- ⑤ Value of median can be determined graphically.

Limitations

- ① It is necessary to arrange the data in order.
- ② It is a positional avg, its value is not determined by each and every observation.
- ③ Median is more affected by sampling fluctuations than the arithmetic mean.
- ④ In some cases cannot be computed exactly as the mean. (For even data it gives approximately mid-point of the two middle terms).

Mode

mode is the value which occurs most frequently in a set of observations and around which the other items of the set cluster densely.

In case of Discrete freqn distribution

mode is the value of x corresponding to the maximum frequency.

Ex x : 1 2 3 4 5 6 7 8
 f : 6 9 11 19 17 15 9 7
 max freq is 19 & corresponding x is 4
 $\Rightarrow$ mode is 4.

* For some cases the above method is not possible

$\hookrightarrow$ if the maximum freq is repeated

$\hookrightarrow$ if the max. freq. occurs in the very beginning or at the end of the distribution

$\hookrightarrow$ if there are irregularities in the distribution

Then mode is determined by the method of grouping.

Ex Calculate the value of mode for the following data.

Marks	10	15	20	25	30	35	40
Freq.	8	12	36	35	28	18	9

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Solⁿ

Grouping Table

X	(f) I	II	III	IV	V	VI
10	8	20	48	56	(83)	(99)
15	12					
20	(36)	(71)				
25	35	(62)	(81)			
30	28			46		
35	18	27		55		
40	9					

Analysis Table

Col no.	20	25	30
I	1	—	—
II	1	1	—
III	—	1	1
IV	—	1	1
V	1	1	—
VI	1	1	1
	4	(5) max	3

$\Rightarrow$ modal value is 25

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Calculation of Mode - Continuous Series

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

where L = Lower limit of the modal class

f_1 = Freq. of the modal class

f_0 = " " " class preceeding the modal class

f_2 = " " " " succeeding the modal class

Ex Calculate the mode

Marks	No. of Student	Marks	No. of Student
Above 0	80	Above 60	28
Above 10	77	Above 70	16
Above 20	72	Above 80	10
Above 30	65	Above 90	8
Above 40	55	Above 100	0
Above 50	43		

Since cumulative freq. is given, therefore first convert into a simple-freq. distribution

Marks	No. of Student	Marks	No. of Student
0-10	3	50-60	15
10-20	5	60-70	12
20-30	7	70-80	6
30-40	10	80-90	2
40-50	12	90-100	8

$\Rightarrow$ modal class is 50-60

$$\text{Mode} = 50 + \frac{15-12}{2 \times 15 - 12 - 10} \times 10 = 55$$

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Ex Determine the modal weight.

Weight	No. of persons	Weight	No. of persons
100-110	4	140-150	33
110-120	6	150-160	17
120-130	20	160-170	8
130-140	32	170-180	2

Grouping method [Since it is difficult to say which is the modal class, hence use grouping method]

Weight	f	II	III	IV	V	VI
100-110	4	10				
110-120	6	16	26	30		
120-130	20	36	56	76	58	
130-140	32	68	100	122	82	85
140-150	33	101	134	151	82	58
150-160	17	50	67	84	27	
160-170	8	10	25	10		
170-180	2					

Col. IV	120-130	Analysis table 130-140	140-150
I	1	1	1
II	1	1	1
III	1	1	1
IV	1	1	1
V	1	1	1
VI	1	1	1
Total	3	5	5

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This is a bi-modal series.

Hence use the formula to determine mode

$$\text{Mode} = 3 \text{ median} - 2 \text{ mean}$$

Weight	m	f	c.f.	$\frac{m-A}{h}$	f.d
100-110	105	4	4	-3	-12
110-120	115	6	10	-2	-12
120-130	125	20	30	-1	-20
130-140	135 ^A	32	62	0	0
140-150	145	33	95	1	33
150-160	155	17	112	2	34
160-170	165	8	120	3	24
170-180	175	2	122	4	8

$N=122$

$\Sigma f.d = 55$

$$\bar{X} = A + \frac{\Sigma f.d}{N} \times h \quad [A=135, N=122, h=10, \Sigma f.d=55]$$

$$= 135 + \frac{55}{122} \times 10 = 139.51$$

$$\text{Median} = \frac{N}{2} \text{th item} = \frac{122}{2} = 61 \text{th item}$$

$$\therefore \text{median class} = 130-140$$

$$\text{Median} = L + \frac{h}{f} \left(\frac{N}{2} - c.f. \right) \quad [L=130, h=10, f=32, N=122, c.f.=30]$$

$$= 130 + \frac{10}{32} (61 - 32) = 139.69$$

$$\therefore \text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$$

$$= 3 \times 139.69 - 2 \times 139.51$$

$$= 140.05$$

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Merits of mode

- Like median, mode is not unduly affected by extreme values.
- Its value can be determined in open-end distribution, without ascertaining the class limits.
- It can be used to describe qualitative phenomenon.
- The value of mode can also be determined graphically.

Limitations

- It can not always be determined. In some cases we may have a bimodal series.
- It is not capable of algebraic manipulations.
- Mode is not based on each & every item of the series.
- While dealing with quantitative data, the disadvantages of the mode outweigh its good features and hence it is seldom used.

Measures of Variability

- Range
- Quartile Deviation or Semi-interquartile Range
- Mean Deviation
- Standard Deviation & Root mean square deviation

Range

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

Range is the difference between two extreme observations of the distribution.

- Since it is based on two extreme observations which themselves are subject to chance fluctuations, therefore, it is not at all a reliable measure of dispersion.

Quartile deviation

The three points which divide the series into four equal parts are called quartiles.

1st quartile, Q_1 , is the value which is exceeded by 25% of the observations and is exceeded by 75% of the observations.

2nd quartile, Q_2 , coincides with median.

3rd quartile, Q_3 , is the pt. which has 75% observations before it and 25% observations after it.

$$\text{Quartile deviation} = Q = \frac{1}{2} (Q_3 - Q_1)$$

where Q_1 & Q_3 are 1st & 3rd quartiles

Quartile deviation is a better measure than the range as it makes use of 50% of the data. But ignores the other 50% of the data, it cannot be regarded as a reliable measure.

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Ex. Calculate median, quartiles

X	0	1	2	3	4	5	6	7	8
f	1	9	26	59	72	52	29	7	1
c.f	1	10	36	95	167	219	248	255	256

Median $\frac{N}{2} = \frac{256}{2} = 128$

c.f. greater than 128 is 167
 $\Rightarrow$ median = 4

$Q_1: \frac{N}{4} = 64$, c.f. greater than 64 is 35
 $\Rightarrow Q_1 = 3$

$Q_3: \frac{3N}{4} = 192$, c.f. greater than 192 is 219
 $\Rightarrow Q_3 = 5$

$\therefore$ quartile deviation = $\frac{1}{2}(Q_3 - Q_1) = \frac{5-3}{2} = 1$

Mean Deviation

If x_i/f_i , $i=1, 2, \dots, n$ is the freq. distribution, then mean deviation from the average A , (Usually mean, median, mode) is given by

$$\frac{1}{N} \sum f_i |x_i - A|, \quad \sum f_i = N$$

where $|x_i - A|$ represents the modulus of the absolute value of the deviation $(x_i - A)$, where the -ve sign is ignored. It is a better measure of dispersion, since it is based on all the observations.

But ignoring the signs of the deviations $(x_i - A)$ creates artificiality and renders it useless for further mathematical treatment.

* Mean deviation is least when taken from median.

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Calculation of mean deviation

$$M.D. = \frac{1}{N} \sum f_i |x_i - A|$$

Ex. Calculate mean deviation from the following series

X	f	c.f	$ x_i - 12 $	f D
10	3	3	2	6
11	12	15	1	12
12	18	33	0	0
13	12	45	1	12
14	3	48	2	6

$N = 48$
 $\sum f|D| = 36$
 $\text{median} = \frac{N+1}{2} \text{ item} = \frac{48+1}{2} = 24.5^{\text{th}} \text{ item}$

$\Rightarrow$ median = 12

mean deviation = $\frac{36}{48} = 0.75$

X	f	fX	$ x - 8 $	f x-8
2	2	4	6	12
4	2	8	4	8
6	4	24	2	8
8	5	40	0	0
10	3	30	2	6
12	2	24	4	8
14	1	14	6	6
16	1	16	8	8

$N = 20$
 $\sum fX = 160$
 $\bar{X} = \frac{\sum fX}{N} = \frac{160}{20} = 8$

$M.D. = \frac{\sum f|D|}{N} = \frac{56}{20} = 2.8$

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Standard deviation (σ)

For the freq. distribution x_i/f_i , $i=1, 2, \dots, n$

$$\sigma = \sqrt{\frac{1}{N} \sum f_i (x_i - \bar{x})^2}$$

(It measures the absolute dispersion or variability of distribution)

where $\bar{x}$ is the arithmetic mean.

Variance

Square of standard deviation is variation-variance.

$$\sigma^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$$

Root Mean Square Deviation

$$S = \sqrt{\frac{1}{N} \sum f_i (x_i - A)^2}$$

where A is any arbitrary no. $\in$

In a frequency distribution where deviations are taken from assumed mean variation may directly be computed as

$$\text{variance} = h^2 \left[\frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \right]$$

where $d = \frac{x_i - A}{h}$

Standard deviation

when deviations are taken from assumed mean

$$\sigma = \sqrt{\frac{\sum d^2}{N} - \left(\frac{\sum d}{N} \right)^2}$$

where $d = x - A$

For discrete series using assumed mean

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N} \right)^2}, \quad d = x - A$$

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Mean deviation - Continuous Series

X	f	c.f	mid	mid-43	f D
0-10	5	5	5	38	190
10-20	8	13	15	28	224
20-30	12	25	25	18	216
30-40	15	40	35	8	120
40-50	20	60	45	2	40
50-60	14	74	55	12	168
60-70	12	86	65	22	264
70-80	6	92	75	32	192

$\sum f = 92$
 $\text{Med} = \text{Size of } \frac{N}{2} = \frac{92}{2} = 46^{\text{th}}$

Median class 40-50

$$\text{Median} = L + \frac{\frac{N}{2} - cf}{f} = 40 + \frac{10}{20} (46 - 40) = 43$$

Mean deviation from median = $\frac{\sum f|D|}{N} = \frac{1414}{92} = 15.37$

Coeff of M.D. = $\frac{M.D.}{\text{Median}} = \frac{15.37}{43} = 0.357$

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Ex. Calculate the mean & standard deviation for the following table

Age in years	No. of members	Mid value x	$d = \frac{x-55}{10}$	fd	fd^2
20-30	3	25	-3	-9	27
30-40	61	35	-2	-122	244
40-50	153	45	-1	-153	153
50-60	153	55	0	0	0
60-70	140	65	1	140	140
70-80	51	75	2	102	204
80-90	2	85	3	6	18
	$\Sigma f = 542$			$\Sigma fd = -15$	$\Sigma fd^2 = 765$

$$\bar{x} = A + h \frac{\Sigma fd}{N} = 55 + 10 \frac{(-15)}{542} = 55 - 0.28 = 54.72$$

$$\sigma^2 = h^2 \left\{ \frac{1}{N} \Sigma fd^2 - \left(\frac{1}{N} \Sigma fd \right)^2 \right\}$$

$$= (10)^2 \left\{ \frac{1}{542} \times 765 - \left(\frac{1}{542} \times (-15) \right)^2 \right\}$$

$$= 100 \times 1.333 = 133.3$$

Standard deviation $\sigma = 11.55$ year

Coefficient of Variation

$$C.V. = 100 \times \frac{\sigma}{\bar{x}}$$

It is a relative measure of dispersion

It is used for comparing the variability of two series.

Ex. From the prices of shares of X & Y below find out which is more stable

	X	Y
	35 54 52 53 56 58 52 50 51 49	108 107 105 105 106 107 104 103 104 101
Sol.	X	Y
	$x - \bar{x}$	$y - \bar{y}$
	$(x - \bar{x})^2$	$(y - \bar{y})^2$
	35	108
	54	107
	52	105
	53	105
	56	106
	58	107
	52	104
	50	103
	51	104
	49	101
	$\Sigma x = 510$	$\Sigma y = 1050$
	$\Sigma (x - \bar{x})^2 = 350$	$\Sigma (y - \bar{y})^2 = 40$

For X:

$$\bar{x} = \frac{\Sigma x}{N} = \frac{510}{10} = 51$$

$$\sigma = \sqrt{\frac{\Sigma (x - \bar{x})^2}{N}} = \sqrt{\frac{350}{10}} = 5.916$$

$$C.V. = 100 \times \frac{\sigma}{\bar{x}} = \frac{5.916}{51} \times 100 = 11.6$$

For Y:

$$\bar{y} = \frac{\Sigma y}{N} = \frac{1050}{10} = 105$$

$$\sigma = \sqrt{\frac{40}{10}} = 2$$

$$C.V. = 100 \times \frac{2}{105} = 1.905$$

$\therefore$ C.V. for Y is less $\Rightarrow$ Y is more stable

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Coefficient of dispersion (C.D.)

These are the coefficients i.e. pure numbers independent of the units of measurement.

(1) C.D. based on range = $\frac{A-B}{A+B}$, where A & B are the greatest & smallest items

(2) C.D. based on quartile deviation

$$C.D. = \frac{(Q_3 - Q_1)/2}{(Q_3 + Q_1)/2} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

(3) C.D. based on mean deviation

$$C.D. = \frac{\text{Mean deviation}}{\text{Average from which it is calculated}}$$

(4) Based on standard deviation

$$C.D. = \frac{\text{S.D.}}{\text{Mean}} = \frac{\sigma}{\bar{x}}$$

Moments

In statistics it describes the various characters of a freq. distribution like Central tendency, Variation, Skewness & Kurtosis.

The r th moment of a Variable x about any point $x=A$, denoted by μ_r'

$$\mu_r' = \frac{1}{N} \sum f_i (x_i - A)^r, \quad \Sigma f_i = N$$

$$= \frac{1}{N} \sum f_i d_i^r, \quad d = x_i - A$$

r th moment about mean $\bar{x}$

$$\mu_r = \frac{1}{N} \sum f_i (x_i - \bar{x})^r = \frac{1}{N} \sum f_i z_i^r$$

$$z_i = x_i - \bar{x}$$

Particular case

$$\mu_0 = \frac{1}{N} \sum f_i (x_i - \bar{x})^0 = \frac{1}{N} \sum f_i = 1$$

$$\mu_1 = \frac{1}{N} \sum f_i (x_i - \bar{x}) = 0, \quad \text{The Algebraic sum of deviations from mean zero.}$$

$$\mu_2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2 = \sigma^2$$

$$\mu_0 = 1, \mu_1 = 0, \mu_2 = \sigma^2$$

4 Central moments

- 1st moment $r=1$, sum of the difference of each observation from sample avg., is zero
- 2nd central moment $r=2$, is Variance
- 3rd $r=3$, is Skewness
- 4th central $r=4$, is Kurtosis.

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Pearson's co-efficient

Karl Pearson defined the following four coefficients based on first four moments about mean

$$\beta_1 = \frac{\mu_3}{\mu_2^{3/2}}$$

$$\gamma_1 = +\sqrt{\beta_1}$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\gamma_2 = \beta_2 - 3$$

Skewness (Lack of symmetry)

A distribution is said to be skewed if

- (i) Mean, median and mode fall at diff. points
Mean $\neq$ median $\neq$ mode
- (ii) Quartiles are not equidistant from median.
- (iii) The curve drawn with the help of the given data is not symmetrical but stretched more to one side than to the other.

Measure of skewness

- (1) $S_k = M - M_d$
- (2) $S_k = M - M_o$

M = mean
M_o = Mode
M_d = Median

$$(3) S_k = (Q_3 - M_d) - (M_d - Q_1)$$

These are all absolute measure of skewness.

For comparing two series we calculate the relative measure, called coefficients of skewness

1) Karl Pearson's coeff. of skewness

$$S_k = \frac{M - M_o}{\sigma}$$

2) Bowley's coeff. of skewness

$$S_k = \frac{(Q_3 - M_d) - (M_d - Q_1)}{(Q_3 - Q_1) + (M_d - Q_1)} = \frac{Q_3 + Q_1 - 2M_d}{Q_3 - Q_1}$$

3) Based on moments

$$S_k = \frac{\sqrt{\beta_1} (\beta_2 + 2)}{2(5\beta_2 - 6\beta_1 - 9)}$$

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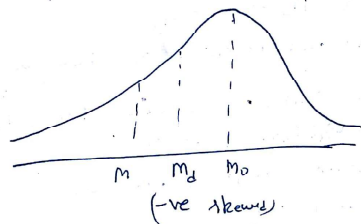
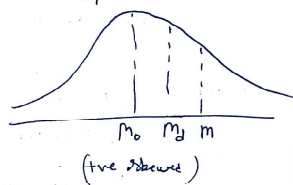
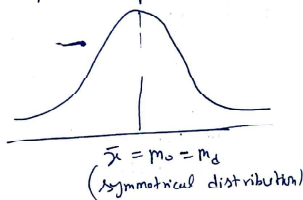
$$S_k = 0, \text{ if } \beta_1 = 0, \text{ or } \beta_2 = -3$$

$$\therefore \beta_2 = \frac{\mu_4}{\mu_2^2} \text{ can not be } -ve$$

$$\Rightarrow S_k = 0 \text{ if } \beta_1 = 0$$

$\therefore$ For a symmetrical distribution $\beta_1 = 0$

Graphical representation



Kurtosis

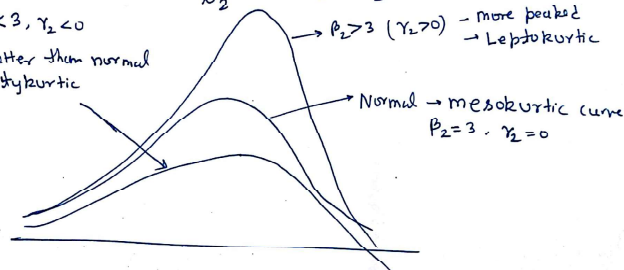
It gives an idea about the flatness or peakedness of the curve.

It is measured by the coeff. β_2

$$\beta_2 = \frac{\mu_4}{\mu_2^2}, \quad \gamma_2 = \beta_2 - 3$$

$$\beta_2 < 3, \gamma_2 < 0$$

Flatter than normal
platykurtic



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